

Syslog-ng Forwarding Guide (pfSense → Wazuh)

Purpose

Suricata generates EVE JSON logs containing structured network security alerts. pfSense's built-in syslog daemon **cannot** forward these logs correctly because it: - Strips JSON structure - Only forwards traditional syslog format - Conflicts with Suricata's file-based logging model

To forward full, unmodified Suricata EVE JSON alerts to the Wazuh SIEM, **syslog-ng** is deployed on pfSense and configured to: - Tail Suricata's eve.json file - Forward raw JSON events to Wazuh via UDP 514 - Replace pfSense's default syslogd to avoid resource conflicts

1. Install syslog-ng

pfSense GUI:

```
System → Package Manager → Available Packages  
Search: syslog-ng  
Install
```

2. Locate Suricata EVE JSON Path

Each Suricata interface has a dedicated folder, e.g.:

```
/var/log/suricata/suricata_vtnet1/eve.json
```

Your interface folder name may differ depending on NIC order.

3. Create syslog-ng Configuration

Edit file:

```
ee /usr/local/etc/syslog-ng.conf
```

Append the following block (replace **WAZUH_IP** accordingly):

```
# --- CUSTOM SURICATA EVE JSON SOURCE ---
source s_suricata_eve {
    file("/var/log/suricata/<INTERFACE>/eve.json" flags(no-parse) follow-
freq(1));
};

# --- DESTINATION: WAZUH MANAGER ---
destination d_wazuh {
    udp("WAZUH_IP" port(514));
};

# --- LOG PATH ---
log {
    source(s_suricata_eve);
    destination(d_wazuh);
};
```

Notes

- **flags(no-parse)** is required to send raw JSON without syslog-ng attempting to interpret it.
- **follow-freq(1)** ensures near real-time updates.

4. Disable pfSense Default syslogd

syslog-ng **cannot** run concurrently with syslogd due to lock on /dev/klog.

Step A — Disable via GUI

Status → System Logs → Settings → Disable writing log files to the local disk.

This stops syslogd.

Step B — Persist Disable Across Reboots

Edit:

```
ee /etc/rc.conf.local
```

Add:

```
syslogd_enable="NO"
```

Step C — Kill Running syslogd Instance

```
kill $(ps ax | grep /usr/sbin/syslogd | grep -v grep | awk '{print $1}')
```

5. Protect syslog-ng Configuration (Critical)

pfSense sometimes rewrites config files when plugins restart. The solution is to make the syslog-ng config immutable.

```
chflags schg /usr/local/etc/syslog-ng.conf  
chown root:wheel /usr/local/etc/syslog-ng.conf  
chmod 644 /usr/local/etc/syslog-ng.conf
```

This prevents deletion or corruption by pfSense GUI processes.

6. Start syslog-ng

```
/usr/local/etc/rc.d/syslog-ng onestart
```

Confirm it loads without conflict.

7. Verify Forwarding to Wazuh

On the Wazuh Manager

Run tcpdump to confirm packet arrival:

```
sudo tcpdump -i eth0 -n -s 0 -A 'udp port 514 and host <PFSENSE_WAN_IP>'
```

Generate test alert:

```
curl http://testmyids.com
```

Expected output: - JSON event appears in tcpdump capture - Event shows in Wazuh dashboard within seconds

8. Troubleshooting

No logs reaching Wazuh

- Confirm syslog-ng service is running
- Ensure pfSense WAN has proper route to SIEM
- Validate correct EVE JSON path
- Temporarily remove immutable flag to re-edit the file

syslogd respawns after reboot

Check: - /etc/rc.conf.local contains `syslogd_enable="NO"` - GUI log settings have disk logging disabled

EVE JSON not updating

- Suricata may not be bound to the correct interface
 - Rule categories may be disabled
-

9. Summary of Responsibilities

syslog-ng is responsible for: - Transporting raw Suricata JSON alerts to Wazuh - Ensuring no loss of detail from pfSense - Replacing pfSense's default logger for compatibility - Providing real-time event delivery for SIEM correlation

This pipeline is essential to achieving the full network visibility required by the Virtual SOC environment.